12. $\lim_{n \to \infty} a_{n+1} = \lim_{n \to \infty} a_n = \frac{1}{3}$

$$\lim_{n \to \infty} a_{n+1} = \lim_{n \to \infty} \frac{2}{3} a_n + \beta$$

$$= \frac{2}{3} \cdot \frac{1}{3} + \beta$$

$$= \frac{1}{3}$$

$$\therefore \frac{2}{9} + \beta = \frac{1}{3}$$

$$\therefore \beta = \frac{1}{9}$$

▨